VOLODYMYR POLISHCHUK

Head of Software & Autonomy | Systems Architect | GPS-Denied UAV Navigation

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SUMMARY

Systems architect and tech lead with 19 years in production engineering: Wix's serverless developer platform, used by about 3,000 developers across 2,000+ microservices; a real-time betting engine taking 1M requests/s at peak at Playtech; and payment-network infrastructure at UkrCard. Since 2026 I have headed software and autonomy at Applied Technology Collective, a Ukrainian UAV developer. I joined as its only software engineer, built GPS-denied navigation, now in production, that runs on a $15 companion computer, with its own flight-controller firmware and simulator, and hired the team. I start from the physics and the measurements, not from frameworks. I build AI-native: I own the architecture, the estimation math and the acceptance tests, and direct AI coding agents as the implementation team.

EXPERIENCE

Applied Technology Collective | Head of Software & Autonomy — 2026 – Present

Ukrainian developer of strike, loitering-munition and cargo UAVs. I joined as the only software engineer, built the software and autonomy function, and hired two engineers. NADIR is in production, SEXTANT is rolling out, and ORRERY is the standalone simulator both are tested in. All three are my own IP, which the company uses. Code walkthrough on request.

Engineering process

- AI-native delivery: I set the architecture, the estimation math, the interfaces and the acceptance criteria; AI coding agents (Claude Code) write the code under my review. Every change is gated by unit and property tests, with replays of recorded flights and simulator flights as acceptance, and I debug from flight logs; the root causes below were found that way. 1,400+ commits in four months across four repositories.

NADIR — position hold for ArduPilot from one downward camera

- Shipped a product, not a demo: a Rust daemon on a Raspberry Pi Zero 2 W turns camera frames into metric body velocity at 20 Hz and a feature-anchored position fix, both fused by ArduPilot's EKF3. Sealed read-only SD image, HTTPS web console, in-field calibration from QGroundControl, and a pre-arm gate that refuses flight until lens, latencies and autopilot parameters check out. 25 releases in 6 weeks.
- Vision pipeline: FAST corners and gyro-seeded pyramidal KLT, de-rotation of each corner at its own rolling-shutter row time, a RANSAC velocity solve whose sigma tracks the true error, and a BRIEF keyframe anchor matched on gravity-levelled corners. In simulation: 2% position error over a 385 m flight; 40 m hovers hold within 0.6 m.
- Tightly coupled visual-inertial odometry (MSCKF, the OpenVINS family) estimating velocity, attitude, IMU biases, camera–IMU time offset and feature depths: 3.6% and 7.0% drift of distance on recorded flights, against OpenVINS's 4.6% and 4.0% on the same data, within a Pi Zero 2 W's frame budget.
- Root-caused instabilities to physics rather than tuning them away: drone rocking to a 50 ms gyro/image timing offset; a cruise scale runaway to a 1–2.5° attitude-levelling bias feeding back through the velocity; heading swings to ArduPilot's backup attitude estimator reading braking as tilt (heading-induced position error 9–76 m down to 1–5.5 m over five flights). Built a field latency bench that measures camera, IMU and link delays on the aircraft.
- Own ArduCopter 4.7.1 build for the SpeedyBee F405 (1 MB flash): enables the EKF3 body-odometry and external-navigation fusion that stock 1 MB builds leave out, fixes a PosHold bug that let stock 4.7 drift downwind after any stick input, and fits in 932 of 976 KB. The simulator runs the same source under a feature-parity check.
- Migrated the daemon from C to Rust as a strangler migration, one module at a time, the drone flyable after every step. Coordinate frames and units became types, closing the class of bug that had cost flights. 38k lines of Rust, 418 unit tests, 41 property-test suites, Miri in CI.

SEXTANT — long-range navigation by terrain geometry

- Absolute position and heading without GPS by matching geometry instead of appearance: the forward camera's skyline against a terrain model gives heading and position, downward motion-stereo gives relief, fused into EKF3 on a lane of their own. Over a real 15 km Carpathian corridor (Copernicus GLO-30 terrain, simulated flight): 39 m RMS and 0.53° heading without a magnetometer, re-locking 900 m after a cloud blackout, where the INS alone drifts past 300 m.
- Prototyped and retired SILO, satellite-image matching with MobileNetV3 embeddings on an edge NPU: appearance changes with snow, foliage and sun, terrain geometry does not. Decisions recorded as ADRs.

ORRERY — standalone simulator for camera-based flight

- A quadcopter simulator in Godot 4, run in lockstep with ArduPilot SITL, in which NADIR and SEXTANT both fly their acceptance tests. Driven entirely by configuration: the scenario file starts any ArduPilot SITL build, stock or custom, and each camera (sensor, lens, mount, frame rate, exposure) and the vehicle's physics come from preset files.
- Cameras are modelled as sensors: rolling shutter, motion blur, auto-exposure and noise, with OV5647, IMX219 and IMX296 presets, over 15 cm orthophoto terrain. Frames carry the real driver's headers and timestamps, so timing bugs reproduce on the desk.

Wix.com | **Principal Software Engineer** / **Architect** 2017 – 2026

- Led Wix's adoption of gRPC and Protobuf as the contract for its internal APIs, the foundation for opening them to third-party developers and the Wix App Market; built the Protobuf linter in the core CI/CD pipeline that every service's API passes, with code and documentation generated from the contract.
- Designed and taught the asynchronous-programming courses and workshops, mandatory for Wix's 1,000+ server engineers, that moved the backend to async code and cut service resource use.
- Interviewed weekly throughout my time at Wix (300+ interviews), at all three stages of the company-wide hiring loop: system design, coding, and the Server Guild deep-dive into a candidate's experience and interests that decides where in the company they fit. Sat on the hiring committees: scored and voted on each candidate and helped shape the new hire's career path.
- Architected the internal serverless platform (App Engine) and built it with a core team of five. It let about 3,000 developers, mostly front-end, ship Node.js services across 2,000+ microservices without touching Kubernetes, service wiring or CI/CD; control plane in Scala and ZIO with typed errors.
- Designed a throughput-based autoscaler beyond what KEDA/HPA offer: it pre-warms capacity before datacenter switches and adjusts CPU and memory live, scaling without downtime.
- Architected the Wix App Market's AI chat agent: retrieval-augmented generation on Vespa, and a bidirectional streaming pipeline from WebSockets to a gRPC server for real-time LLM responses.
- Domains: automated SSL certificate issuance and renewal for 8M+ domains with no manual steps; architected Wix's ICANN-accredited registrar platform and an in-house DNS server; led the migration of the legacy Java domain-management monolith to microservices.

Playtech | **Software Architect** / **Tech Lead** 2012 – 2017

- Tech lead of a team of five engineers (a team leader handled management): I set the design, the technical direction and the goals, and ran the team's internal communication. Led the sports-betting platform's migration from a Java EE monolith to a reactive Scala architecture (Scalaz, algebraic domain modeling).
- Real-time streaming held 60M requests/min peaks during the FIFA World Cup over WebSockets and raw sockets, with back-pressured Akka Streams keeping memory bounded under load.
- Cut the bare-metal footprint to a highly available 3-node minimum, with New Relic observability.
- Regulation ruled out the cloud, so I designed a Dynamo-style, eventually consistent key-value store on LevelDB over bare-metal disks, with Akka Cluster gossip for membership and state machines for replica logic; reads are served locally, with no network round-trip.

UkrCard | **Systems Architect** 2007 – 2012

- Led the move of every ATM, POS and self-service terminal on the payment network from wired X.25 to wireless TCP/IP, with end-to-end VPN and SSL encryption (2008–2012).
- Fraud-detection assistant (Master's project, 2007–2008): a neural network written from scratch in Java, ensembled with a human-in-the-loop meta-model that learned from operator feedback to cut false positives under severe class imbalance.

SKILLS

Autonomy: visual and visual-inertial odometry (MSCKF), EKF sensor fusion, camera calibration and timing, terrain-referenced navigation, ArduPilot (EKF3, firmware, SITL), MAVLink.
Languages: Rust, C, Scala (ZIO, Akka, Scalaz), Java, Python, Node.js.
Systems: embedded Linux on ARM, real-time pipelines, Kubernetes, gRPC/Protobuf, distributed storage, CI/CD, Godot.
AI: directing AI coding agents, RAG, neural networks from first principles.

EDUCATION

M.Sc. in Cybernetics | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics. Thesis: Ensuring the Dependability of Transaction Classification in a Payment System.

B.Sc. in Applied Mathematics | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics

MILITARY & OTHER QUALIFICATIONS

Armed Forces of Ukraine (reserve): Junior Lieutenant, military specialty 708200: operation, tactical deployment and maintenance of air-defense radar systems.
IYT Skipper (International Yacht Training): practical navigation.
Interests: world models for robot navigation (V-JEPA, Genesis); basal cognition (Michael Levin); foundations of physics.